

The Double-Ended Helix and Projection Primacy: A Formally Verified Three-Dimensional Carrier for Dirichlet L -Functions, with the Generalized Riemann Hypothesis as the Completeness of its Projection

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Abstract

We introduce a formally specified three-dimensional, double-ended *Frobenius-dual helix carrier* for Dirichlet L -function phasors. The main point is not numerical agreement with known zeros, but a Lean-verified package of geometric, analytic, and spectral statements. Uniform arclength placement on the growing-radius helix forces the area law $r_n \sim C\sqrt{n}$ (`windIntegerSite_radius_sq_tendsto`); consequently $n^{-1/2}$ is the unique reciprocal scale for a stable fiber (`sigma_half_is_scale_critical`, `scaleBalanced_iff`). Thus the abscissa $\operatorname{Re} s = \frac{1}{2}$ is derived from the carrier's scale geometry before any zero-location statement enters.

The analytic fiber is equally explicit. For a non-principal Dirichlet character χ , the phasor partial sums $\sum_{n \leq N} \chi(n)n^{-s}$ go all the way into the half-plane $\operatorname{Re} s > 0$ and converge exactly to $L(s, \chi)$ (`dirichlet_strip_tendsto_LFunction`), not merely in the absolutely convergent region. The proof formalizes bounded character sums, Abel summation, locally uniform tail summability, and an identity-theorem identification with Mathlib's L -function. For the principal case, the eta-regularized channel converges exactly to $(1 - 2^{1-s})\zeta(s)$ on $\operatorname{Re} s > 0$, $s \neq 1$ (`eta_strip_tendsto`); on the critical line the correction factor is nonzero, so vanishing of the regularized fiber is exactly vanishing of ζ (`continuous_model_zeta`, `FullFiber_eq_zero_iff_zeta`). There is no cutoff error or asymptotic remainder in these identities.

The three-dimensional lift separates the charged spinning channels from the neutral mass channel while preserving the L -series readout (`phasor3D_plane_tsum`, `phasor3D_mag3`). The carrier's angular winding is built from prime angles by the fundamental theorem of arithmetic; the logarithm appears only in the external Mellin bridge $w \leftrightarrow n^{it}$ (`wind_glog_eq_cpow`). Prime multiplication acts by a Frobenius similitude $w \mapsto \sqrt{m}\operatorname{wind}(m)w$; conjugate chiral eigenphases form a determinant-one unitary transverse block (`frobeniusSimilitude_mul`, `frobeniusBlock_unitary`). The local vanishing mechanism is also spectral: on the line the finite fiber is the ℓ^2 inner product of the sampled eigenwave $\psi_\gamma(t) = e^{i\gamma t}$ with the arithmetic data, and vanishing is exactly orthogonality (`fiberCarrierFinite_eq_inner`, `fiberCarrierFinite_eq_zero_iff_orthogonal`).

The harmonic-pencil layer packages the same event algebraically. The exact $\pi/3$ cell scaling is not cosmetic: it is the nonzero six-cell normalization that turns the signed channel closure into $(\pi/3)L(\frac{1}{2} + i\log Z, \chi)$, so the harmonic cancellation event is exactly the L -zero event (`Ucell_ne_zero`, `cell_normalized_scalar_exact`, `scalar_zero_iff_Lzero`). The signed and unsigned channels assemble a 2×2 pencil whose determinant, and equivalently its Gram rank-drop, vanishes exactly at the corresponding critical-line L -zero (`harmonic_pencil_det_zero_iff_Lzero`, `gramH_rank_drop_iff_Lzero`). More generally,

the no-log-trap helix-height pencil has Hermitian positive Gram form and real generalized eigenheights as Rayleigh quotients (`helix.eigenheight_real`, `helix.eigenheight_rayleigh`). A real source-height readout lands on the critical line, and every point on that line has such a readout (`admissible_iff_re_half`); projection primacy is the separate completeness principle that the real-height readout exhausts all nontrivial zeros (`ProjectionPrimacy`, `projectionPrimacy_iff_exhausts`).

Every named statement below is formalized in Lean 4 with Mathlib and checked by the kernel. The project builds with no `sorry`, `admit`, or construction-specific axiom; the axiom footprint of the named theorems is `{propext, Classical.choice, Quot.sound}`.

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1 Main formalized contributions

The formal development proves four independent pieces which are easily blurred if the helix picture is read only as a visualization.

- (1) *Scale geometry.* The $\sqrt{n}$ radius is not stipulated. Integers are placed at fixed arclength spacing on the unwound line and then wound onto the helix; the resulting cylindrical radius satisfies $r_n^2/n \rightarrow r(\pi/3)/\pi$ (Theorem 3.9). From this, the unique scale-balanced exponent is $\sigma = \frac{1}{2}$ (Theorem 3.11).
- (2) *Exact analytic accumulation.* For non-principal characters, the phasor channel goes into all of $\text{Re } s > 0$ and converges exactly to $L(s, \chi)$ (Theorem 7.1); for ζ , the eta channel reaches the same half-plane and identifies zero events on the critical line exactly (Theorems 7.2, 7.3, 7.5). This is the analytic core of the model: the fiber has an exact value in the strip, not a finite-window approximation.
- (3) *Spectral and pencil readouts.* The local on-line vanishing event is an eigenwave-data orthogonality identity (Proposition 12.8); the harmonic pencil and its Gram form have rank-drop exactly at the same L -zero event (Theorems 16.3, 16.4). The normalizing factor is the exact $\pi/3$ cell unit, so the harmonic cancellation is an exact scalar multiple of the L -fiber rather than a fitted calibration. The finite helix-height pencil has the self-adjoint positivity mechanism that forces real eigenheights.
- (4) *Admissibility boundary.* A real source-height readout reaches exactly the critical line, and every point of the critical line is represented by such a height (Theorem 16.5). Completeness of this readout is named separately as projection primacy; it is not used to prove the unconditional on-line identities.

The rest of the paper expands these four points and records the Lean identifiers for each theorem.

2 Objects and notation

Let χ be a Dirichlet character modulo q and $L(s, \chi)$ the associated L -function; ζ is the Riemann zeta function and $\Lambda(s) = \text{completedRiemannZeta}(s)$ its completed form, normalized so that $\Lambda(1-s) = \Lambda(s)$. The *critical line* is the set $s = \frac{1}{2} + iy$, $y \in \mathbb{R}$. We write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ for the unit circle (`Mathlib.Circle`) and $\bar{\cdot}$ for complex conjugation (`starRingEnd` $\mathbb{C}$). For an arithmetic weight χ the *phasor* attached to the integer $n \geq 1$ on the vertical line $s = \sigma + iy$ is $\chi(n) n^{-s}$.

The *abscissa* $\frac{1}{2}$. Every $\frac{1}{2}$ below is one of two *derived* constants, neither of which assumes where the zeros lie. As the carrier's *scale-critical exponent* it is the unique σ at which the phasor amplitude $n^{-\sigma}$ balances the emergent area-law radius $r_n \sim \sqrt{n}$ (Theorem 3.11); this fixes the amplitude $n^{-1/2}$ and the abscissa of the line the fiber rides, from the carrier geometry alone. As the *reflection axis* $\text{Re } s = \frac{1}{2}$ it is the fixed line of $s \mapsto 1-s$, on which $\bar{s} = 1-s$; with the functional equation $\Lambda(1-s) = \Lambda(s)$ this is what makes the completed object real on the line (Theorem 9.3) and pairs the two chiralities (Theorem 11.3). The Riemann Hypothesis — that the *zeros* ρ satisfy $\text{Re } \rho = \frac{1}{2}$ — is a third, distinct statement, never assumed here: the on-line results below are either unconditional in y or conditional on a zero being present at the stated height, and they assert nothing about off-line zeros.

The classical analytic theory of ζ and Dirichlet L -functions — the Euler product, the functional equation, and the von Mangoldt explicit formula [1, 2, 3, 4, 5, 7] — enters here through the corresponding Mathlib lemmas.

All named results carry the identifier of the corresponding Lean declaration, e.g. `frobenius_conjugate_det_one`; Section 18 records the verification.

3 The $\pi/3$ helix carrier

3.1 The space curve and its arclength

Definition 3.1 (Carrier curve; `Geometry.helix`, `helixVel`, `speed`). For pitch p and radial rate r , the carrier is the growing-radius helix

$$\gamma(k) = (r k \cos 2\pi k, r k \sin 2\pi k, p k), \quad k \in \mathbb{R}.$$

Its velocity is $\gamma'(k) = (r \cos 2\pi k - 2\pi r k \sin 2\pi k, r \sin 2\pi k + 2\pi r k \cos 2\pi k, p)$, and the squared speed is

$$\|\gamma'(k)\|^2 = p^2 + r^2 + (2\pi r k)^2.$$

The climber is $k(y) = e^y/p$, so the height is $z = p k(y) = e^y$ and the cylindrical radius is $R(y) = r k(y) = (r/p)e^y$.

Remark 3.2 (Constant height law: $\Delta z = 1$ per integer n , not per phasor; `Geometry.heightLaw`, `heightLaw_step`, `resonance_height`, `activeFraction_eq_totient`, `carrierRadius_growth_div_sqrt_tendsto`). Discretely, the carrier assigns integer n the height $z_n = n$: a constant $\Delta z = 1$ per integer n (`heightLaw_step`: $\text{heightLaw}(n+1) - \text{heightLaw}(n) = 1$)—every slot, whether or not its arithmetic weight $\chi(n)$ is nonzero. A neutral slot ($\chi(n) = 0$) carries no arrow yet still occupies its place and still advances the height. Counting instead per active phasor would climb by the active fraction $\varphi(q)/q$ per integer (`activeFraction_eq_totient`: only the $\varphi(q)$ unit residues carry a spinning arrow) and land the resonance at $(\varphi(q)/q)e^\gamma$, short by exactly that fraction. Since phasor n resonates at $y = \log n$, integer n sits at $z_n = n = e^{\log n}$ —the height of its own resonant frequency (`resonance_height`)—so a zero at frequency γ (resonant term $n^* = e^\gamma$) lands at height

$z = e^\gamma$, for every L -function (it rides only on $\text{spin} = \log n$; the located on-line vanishings are exactly the carrier's resonant heights, `online_zero_eq_carrierZeros`, each at a unique height, `online_zero_unique_height`). This is the discrete form of the climber height $z = pk(y) = e^y$ above. The radial gap per turn is L -dependent, $g = e^{(q \bmod 1)}$ ($\eta e^2, \chi_3 e^3, \dots$), so the area-law radius specializes to $r_n/\sqrt{n} \rightarrow \sqrt{e^q/3}$ (`carrierRadius-growth-div-sqrt-tendsto`): the same $\sqrt{n}$ shape for every L , only the constant differs — consistent with the gauge-independence of the scale-critical exponent (Theorem 3.11).

Definition 3.3 (Arclength; `Geometry.arclength`, `arclengthClosed`). The arclength of γ from 0 to k is $S(k; p, r) = \int_0^k \sqrt{p^2 + r^2 + (2\pi r t)^2} dt$.

Theorem 3.4 (Closed-form arclength; `arclength_closed_form`, `arclength_r_zero`). For $r > 0$,

$$S(k; p, r) = \frac{k}{2} \sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2} + \frac{p^2 + r^2}{4\pi r} \operatorname{arsinh}\left(\frac{2\pi r k}{\sqrt{p^2 + r^2}}\right),$$

and $S(k; p, 0) = pk$ for $p \geq 0$.

Proof sketch. Differentiate the stated antiderivative and check it equals the integrand $\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2}$ pointwise, then apply the fundamental theorem of calculus on $[0, k]$; the constant-pitch case is $\int_0^k p dt$. $\square$

3.2 Uniform integer placement and the area law

Definition 3.5 (Spacing, index, integer angle; `Geometry.Delta`, `Nindex`, `spinAngle`). The fixed spacing is $\Delta = \pi/3$. The continuous geometric index is $N(y) = S(k(y); p, r)/\Delta$, so $N(y) = (3/\pi) S(k(y); p, r)$ (`Nindex_eq`). The integer angular coordinate is $s_n = n \cdot (\pi/3)$.

Theorem 3.6 (Uniform $\pi/3$ spacing; `carrier_spacing`). For every $n \in \mathbb{N}$, $s_{n+1} - s_n = \pi/3$.

Theorem 3.7 (Area-law bound; `arclengthClosed_lower_bound`, `arclengthClosed_upper_bound`). For $r > 0$ and $k \geq 0$,

$$\pi r k^2 \leq S(k; p, r) \leq \pi r k^2 + 2k\sqrt{p^2 + r^2}.$$

Definition 3.8 (Wound integer placement; `Geometry.windParameter`, `windIntegerSite`, `exists_unique_windParameter`). Integer n sits at arclength $s_n = n\Delta$ on the unwound line (Definition 3.5). Winding the line onto the helix places it at the unique parameter $k_n \geq 0$ with $S(k_n; p, r) = n\Delta$ (`exists_unique_windParameter`; the map $s \mapsto k_n$ is `windParameter`). The wound site is `windIntegerSite(n) = \gamma(k_n)`, with cylindrical radius $r_n = r k_n$ (`windIntegerSite_cyl_radius`).

Theorem 3.9 (Emergent radius / area law; `emergent_radius_sq_tendsto`, `windIntegerSite_radius_sq_tendsto`). For the wound placement $S(k_n; p, r) = n\Delta$ of Definition 3.8,

$$\frac{r_n^2}{n} = \frac{(r k_n)^2}{n} \xrightarrow{n \rightarrow \infty} \frac{r\Delta}{\pi}, \quad \text{equivalently} \quad r_n \sim \sqrt{\frac{r\Delta}{\pi}} \sqrt{n},$$

so the enclosed area $\pi r_n^2 \sim r\Delta n$ is linear in n . The factor $\sqrt{n}$ is thus derived from the arclength geometry: the bound of Theorem 3.7 forces $S(k) \propto k^2$, hence $k_n \propto \sqrt{n}$ and $r_n = r k_n \propto \sqrt{n}$ — it is not posited.

Remark 3.10 (Unit gauge; `windIntegerSiteRadiusSqTendstoUnitGauge`). The area-law constant is $r\Delta/\pi$; with the native spacing $\Delta = \pi/3$ it equals $r/3$, so $r_n \sim \sqrt{n}$ exactly in the gauge $r\Delta = \pi$, i.e. $r = 3$ (`windIntegerSiteRadiusSqTendstoUnitGauge`); in the alternative gauge $r = 1$ the constant is $1/\sqrt{3}$. In this unit gauge the carrier point $\text{helixPt}(n) = \sqrt{n} \cdot \text{wind}(n)$ (Definition 4.1) has modulus $\|\text{helixPt}(n)\| = \sqrt{n}$ (`HelixLogFree.normHelixPt`), matching the emergent radius r_n asymptotically (`helixPtRadiusMatchesAreaLaw`: the ratio $\|\text{helixPt}(n)\|/r_n \rightarrow 1$ in this gauge); the winding contributes unit modulus (Theorem 4.2).

3.3 The scale-critical exponent

The area law gives more than the radius $\sqrt{n}$: it pins the critical exponent. Write $r_n = \text{carrierRadius}(n)$ for the wound radius of Definition 3.8 (`Geometry.carrierRadius`), and pair it with a candidate weight $n^{-\sigma}$, $\sigma \in \mathbb{R}$.

Theorem 3.11 (The critical exponent is scale-critical; `sigmaHalfIsScaleCritical`). For $\sigma \in \mathbb{R}$, the scale-balance product $n^{-\sigma} r_n$ has a positive limit if and only if $\sigma = \frac{1}{2}$:

$$(\exists L > 0 : n^{-\sigma} r_n \xrightarrow{n \rightarrow \infty} L) \iff \sigma = \frac{1}{2}.$$

Concretely, $n^{-\sigma} r_n \rightarrow 0$ for $\sigma > \frac{1}{2}$ (the weight decays faster than the radius grows), $n^{-\sigma} r_n \rightarrow +\infty$ for $\sigma < \frac{1}{2}$ (the weight outruns the radius), and $n^{-\sigma} r_n \rightarrow \sqrt{r\Delta/\pi} > 0$ for $\sigma = \frac{1}{2}$. Thus $\frac{1}{2}$ is the unique exponent at which the weight reciprocates the emergent radius. It is gauge-independent: the gauge $r\Delta$ sets only the value of the limit, never which σ is critical.

Proof sketch. By Theorem 3.9, $r_n/\sqrt{n} \rightarrow \sqrt{r\Delta/\pi} =: c > 0$. Factor $n^{-\sigma} r_n = (r_n/\sqrt{n}) \cdot n^{1/2-\sigma}$; the first factor tends to c , and the second tends to 0, to 1, or to $+\infty$ according as $\sigma > \frac{1}{2}$, $\sigma = \frac{1}{2}$, or $\sigma < \frac{1}{2}$. Only $\sigma = \frac{1}{2}$ yields a positive finite limit; uniqueness of limits closes both directions. $\square$

Remark 3.12. This is the geometric origin of the critical line: the $\frac{1}{2}$ of $\text{Re } s = \frac{1}{2}$ is the $\frac{1}{2}$ of $r_n \sim \sqrt{n} = n^{1/2}$. It locates the line on which the phasor weights are scale-normalized to $O(1)$, so that the accumulation is governed by the winding (the phase cancellation) rather than the amplitude — which is exactly the mechanism by which the vanishing points arise (Section 7). It is a statement about the carrier alone; that the zeros of ζ and L then lie on this line is the separate analytic content of the functional equation (Theorem 11.3), not of the area law.

3.4 The mod-6 bucket structure

Because $6 \cdot (\pi/3) = 2\pi$, the integer angle s_n is 6-periodic on the circle.

Theorem 3.13 (Bucket periodicity and counts; `spinPhasorPeriod6`, `bucketCount`). $\exp(i s_{n+6}) = \exp(i s_n)$ for all n , and for $a < 6$ exactly N of the integers in $[0, 6N)$ satisfy $n \equiv a \pmod{6}$. Thus the integers collapse onto six angular directions, each receiving an equal share.

Remark 3.14 (Sixth roots of unity; Eisenstein integers). The integer-angle phasor $\exp(i s_n) = \exp(in\pi/3)$ takes values in the sixth roots of unity $\mu_6 = \{e^{ik\pi/3} : 0 \leq k < 6\}$, cycling through them as n runs mod 6; the primitive sixth root $\zeta = e^{i\pi/3}$ ($\zeta^2 = \zeta - 1$, the cyclotomic relation Φ_6) generates μ_6 . The ring $\mathbb{Z}[\zeta]$ is the ring of Eisenstein integers [8] — a hexagonal (triangular) lattice whose unit group is exactly μ_6 . So the $\pi/3$ gauge is the Eisenstein symmetry, and the six buckets of Theorem 3.13 are its units.

Theorem 3.15 (Exact finite monodromy of the $\pi/3$ gauge; `pi3.finite_monodromy`, `unit_not_finite_monodromy`, `circleMonodromy_iff_rat`). Read the carrier phase after k cells modulo a turn, $C_H(k) = kH \pmod{2\pi}$; a cell unit H has exact finite circle monodromy when the carrier returns to its origin after some positive number of cells, i.e. $kH = 2\pi m$ for integers $k \geq 1$, $m \geq 1$. The native gauge $H = \pi/3$ closes exactly after six cells, $6 \cdot (\pi/3) = 2\pi$ ($k = 6$, $m = 1$; `pi3.finite_monodromy`), whereas the unit gauge $H = 1$ never closes (`unit_not_finite_monodromy`: it would force $\pi \in \mathbb{Q}$). In general H has exact finite monodromy iff $H/(2\pi) \in \mathbb{Q}_{>0}$ (`circleMonodromy_iff_rat`). Thus $\pi/3$ is distinguished not only as the Eisenstein unit angle (Remark 3.14) but as a finite-monodromy gauge — the exact closing condition underlying the six buckets of Theorem 3.13.

Remark 3.16 (A closed carrier for an open winding). The exactness of this closure is a genuine structural distinction from the object the carrier serves. The analytic phasor winding $n^{-it} = e^{-it \log n}$ rides incommensurate frequencies — the $\log p$ are $\mathbb{Q}$ -linearly independent — so it never returns: the standard presentation of L has no closed angular structure. The $\pi/3$ carrier, by contrast, closes exactly after six cells, and only the rational-angle gauges close at all (Theorem 3.15). The carrier thus supplies a closed, finite, commensurate angular home — the Eisenstein/ μ_6 hexagonal lattice (Remark 3.14) — for an arithmetic that has none in its analytic form. (The cell enters the readout as a unit-modulus factor, divided out to recover the L -value in Section 7.)

4 The log-free winding and the bridge

4.1 The fundamental-theorem-of-arithmetic winding

Definition 4.1 (Winding angle and winding; `HelixLogFree.windAngle`, `wind`, `helixPt`). Fix a prime-angle assignment θ (in Lean, a function $\theta : \mathbb{N} \rightarrow \mathbb{R}$; only its values on primes enter). The winding angle is the completely additive extension read off the prime factorization,

$$\Theta(n) = \sum_{p^e \parallel n} e \theta(p) = \mathbf{n.factorization.sum}(\lambda p e. e \cdot \theta(p)),$$

and the winding is $\text{wind}(n) = \exp(i \Theta(n)) \in \mathbb{T}$. The carrier point is $\text{helixPt}(n) = \sqrt{n} \cdot \text{wind}(n)$. No logarithm occurs in Θ .

Theorem 4.2 (Multiplicative character; `windAngle_mul`, `wind_mul`, `wind_one`). For $m, n \neq 0$, $\Theta(mn) = \Theta(m) + \Theta(n)$ and $\text{wind}(mn) = \text{wind}(m) \text{wind}(n)$, with $\text{wind}(1) = 1$. Thus $\text{wind} : \mathbb{N} \rightarrow \mathbb{T}$ is a completely multiplicative character [6].

Proof sketch. Θ is a sum over `Nat.factorization`; additivity is `Nat.factorization_mul` (the fundamental theorem of arithmetic). Exponentiating turns additivity into multiplicativity on $\mathbb{T}$ via `Circle.exp_add`. $\square$

4.2 The bridge $\text{wind } n \leftrightarrow n^{it}$

The single place a logarithm enters is the external identification of the geometric winding with the analytic phasor.

Theorem 4.3 (Logarithm is FTA-additive; `real_log_eq_factorization_sum`). For $n \neq 0$, $\log n = \sum_{p^e \parallel n} e \log p$.

Theorem 4.4 (The bridge; `windAngle_glog`, `wind_glog_eq_cpow`). *With the prime-angle assignment $\theta(p) = \gamma \log p$, one has $\Theta(n) = \gamma \log n$ and hence*

$$\text{wind}(n) = n^{i\gamma} \quad (n \geq 1).$$

Proof sketch. Substitute Theorem 4.3 into Θ to get $\Theta(n) = \gamma \log n$; then $\exp(i\gamma \log n) = n^{i\gamma}$ by the principal-branch definition of $n^{i\gamma}$ for the positive real base n . $\square$

5 The phasor fiber and its accumulation

5.1 Phasor decomposition

Theorem 5.1 (Vertical-line phasor and magnitude; `cpow_vertical_line_phasor`, `norm_cpow_vertical_line`). *For a positive real base x and $s = \sigma + iy$,*

$$x^{-s} = x^{-\sigma} \exp(-(y \log x) i), \quad |x^{-s}| = x^{-\sigma}.$$

In particular, on the critical line $n^{-(1/2+iy)} = n^{-1/2} \exp(-(y \log n) i)$ with magnitude $n^{-1/2}$ for every y (`cpow_critical_line`, `norm_cpow_critical_line`).

Definition 5.2 (The spin; `LFunctionPhasor.mellinSpin`). *The spin of the integer n at ordinate y is the unit-modulus factor $\text{spin}(y, n) = \exp(-(y \log n) i)$.*

Theorem 5.3 (The fiber rides the carrier; `fiber_rides_carrier`). *Write $\text{wind}_{-t \log}$ for the winding of Definition 4.1 with the assignment $\theta(p) = -t \log p$, so $\text{wind}_{-t \log}(n) = n^{-it}$ by the bridge (Theorem 4.4). For $n \geq 1$, the carrier magnitude $n^{-1/2}$ — the reciprocal of the emergent radius $\sqrt{n}$ — times this winding is the Dirichlet phasor:*

$$n^{-1/2} \cdot \text{wind}_{-t \log}(n) = n^{-(1/2+it)}.$$

Remark 5.4 (The fiber's exponent is scale-critical; `critical_exponent_is_scale_critical`). *The exponent $\frac{1}{2}$ here is not a chosen constant. By Theorem 3.11 — re-exported at the unit-gauge carrier $r = 3$ as `critical_exponent_is_scale_critical` — it is the unique σ for which the fiber amplitude $n^{-\sigma}$ scale-balances the area-law radius r_n . So the critical line $\text{Re } s = \frac{1}{2}$ that the fiber rides is the scale-balance locus of the carrier, forced by the geometry rather than inserted.*

5.2 Accumulation to L

Definition 5.5 (Finite phasor carrier; `DirichletPhasorCarrier.finiteCarrier`, `phasorTerm`). *$G_{\chi, N}(s) = \sum_{n < N} \chi(n) n^{-s}$, the partial sums of the carrier-riding phasors.*

Theorem 5.6 (Accumulation on $\text{Re } s > 1$; `fiber_accumulates_to_L`, `finiteCarrier_tendsto_LFunction`). *For every Dirichlet character χ and $\text{Re } s > 1$, $G_{\chi, N}(s) \rightarrow L(s, \chi)$ as $N \rightarrow \infty$.*

6 The three-channel three-dimensional phasor

The planar phasor $\chi(n) n^{-\sigma} \exp(-(y \log n) i)$ of Section 5 is a 2D arrow whose length is $\|\chi(n)\| n^{-\sigma}$; on the *neutral* channel $\chi(n) = 0$ it collapses to 0 and the magnitude $n^{-\sigma}$ is lost. The carrier, by contrast, places *every* slot n (Remark 3.2), neutral or not. The reconciliation lifts each term to a genuine vector in $\mathbb{R}^3 = (\text{spin plane } \mathbb{R}^2) \times (\text{mass axis } \mathbb{R})$, modeled as $\mathbb{C} \times \mathbb{R}$.

Definition 6.1 (3D phasor and its three channels; `Phasor3D.phasor3D, channel`).

$$\text{phasor3D}(\chi, \sigma, y, n) = \left(\underbrace{\chi(n) n^{-\sigma} \exp(-(y \log n)i)}_{\text{spin plane}}, \underbrace{(1 - \|\chi(n)\|) n^{-\sigma}}_{\text{mass axis}} \right).$$

The charged channels $\chi(n) = \pm 1$ place the rotating arrow $\pm n^{-\sigma} \exp(-(y \log n)i)$ in the spin plane with zero mass; the neutral channel $\chi(n) = 0$ places the real amplitude $n^{-\sigma}$ on the mass axis with no spin.

Theorem 6.2 (The spin plane carries L unchanged; `phasor3D_plane_tsum`). The spin-plane (complex) coordinate of `phasor3D` is exactly the planar phasor of Section 5, so its accumulation is the same Dirichlet L -series. The 3D lift changes nothing in the analytic representation — it only records what the plane discards.

Theorem 6.3 (Magnitude conservation across all channels; `phasor3D_mag3`). For a quadratic character the full $\mathbb{R}^3$ Euclidean length of every term is exactly $n^{-\sigma}$, on all three channels including the neutral one — where the planar length $\|\chi(n)\| n^{-\sigma}$ vanished. The magnitude lost from the plane is precisely the mass stored on the axis.

Theorem 6.4 (Standing mass vs. travelling spin; `phasor3D_neutral_no_spin, phasor3D_charged_spin, resonantHeight_eq_exp_mellinReadoutFreq`). The neutral term is independent of the frequency y (`phasor3D_neutral_no_spin`): a standing (resonant) mode, not a travelling one, storing positive amplitude $n^{-\sigma}$ on the mass axis (`phasor3D_neutral_mass_pos`) while contributing 0 to the spin-plane L -series (`phasor3D_neutral_plane_zero`). The charged terms rotate at the readout rate $\log n$ as y advances (`phasor3D_charged_spin`) — the Mellin/readout frequency, distinct from the geometric carrier spin $n \cdot (\pi/3)$. The standing mass resonates at this readout frequency $\text{mellinReadoutFreq}(n) = \log n$ and height $\text{resonantHeight}(n) = n = e^{\log n}$ (`resonantHeight_eq_exp_mellinReadoutFreq`), matching the carrier height law $z_n = n = e^{y_n}$ of Remark 3.2.

Remark 6.5 (Three channels, one decomposition; `phasor3D_three_channel_form`). Every term lies in exactly one channel — positive ($\chi(n) = +1$), negative ($\chi(n) = -1$), neutral ($\chi(n) = 0$) — and is determined by it (`phasor3D_three_channel_form`). The signed pair (positive/negative) is the spinning content the L -series sees; the neutral channel is the mass the planar model omits. This positive/negative/neutral split is the channel structure built into the harmonic pencil of Section 16.

7 Strip extension: exact accumulation into $\text{Re } s > 0$

The fiber does not merely approximate the analytic object near the critical line. After Abel summation it has an exact limit throughout the half-plane $\text{Re } s > 0$: for non-principal characters that limit is $L(s, \chi)$ itself, and for the principal channel it is the exact eta-regularized zeta value. Thus the critical-line readout is reached from a proved strip identity, not from a finite truncation.

Theorem 7.1 (Dirichlet strip extension; `dirichlet_strip_tendsto_LFunction`). For a Dirichlet character $\chi \neq 1$ modulo q and $\text{Re } s > 0$,

$$\sum_{n < N} \chi(n) n^{-s} \xrightarrow{N \rightarrow \infty} L(s, \chi).$$

The convergence target is exactly Mathlib's Dirichlet L -function, not an auxiliary regularization.

Theorem 7.2 (Eta strip extension; `eta_strip_tendsto`). For $\operatorname{Re} s > 0$ and $s \neq 1$,

$$\sum_{n < N} (-1)^{n+1} n^{-s} \xrightarrow{N \rightarrow \infty} (1 - 2^{1-s}) \zeta(s).$$

Proof sketch. Bucket cancellation gives a uniform bound on the character sums $\sum_{n < N} \chi(n)$; Abel summation by parts [6, 3] then yields convergence of $\sum \chi(n) n^{-s}$ on $\operatorname{Re} s > 0$, and the limit agrees with $L(s, \chi)$ on $\operatorname{Re} s > 1$ and hence everywhere on the connected strip by the identity theorem. For the alternating weight the factor $1 - 2^{1-s}$ is the even-term correction relating η and ζ . $\square$

Theorem 7.3 (Continuous reach to the line; `continuous_model_zeta`). For every $y \in \mathbb{R}$, at $s = \frac{1}{2} + iy$,

$$\sum_{n < N} (-1)^{n+1} n^{-s} \longrightarrow (1 - 2^{1-s}) \zeta(\tfrac{1}{2} + iy),$$

and the limit is 0 if and only if $\zeta(\frac{1}{2} + iy) = 0$.

Proof sketch. Theorem 7.2 at $s = \frac{1}{2} + iy$ (using $\operatorname{Re} s = \frac{1}{2} > 0$ and $s \neq 1$). On the line the factor $1 - 2^{1-s}$ is nonzero (`correction_factor_ne_zero_on_critical_line`), so the product vanishes exactly when $\zeta(\frac{1}{2} + iy)$ does (`etaTrivial_eq_zero_iff`). $\square$

7.1 Per-height convergence: the cancellation is in the fiber, not the carrier

The accumulation converges height-by-height, and its vanishing is a property of the *fiber*, never of the carrier. The two objects are distinct. The carrier $\operatorname{helixPt}(n) = \sqrt{n} \operatorname{wind}(n)$ has modulus exactly $\sqrt{n}$, independent of the ordinate y and the character χ (`carrier_norm_height_independent`), and never vanishes off the origin (`carrier_ne_zero`): it is a no-radial-drift source. The fiber paired against it is the unit-modulus spin (`fiber_spin_unimodular`) times the arithmetic datum $\chi(n) n^{-1/2}$; attaching it only *rotates* the carrier and cannot alter the $\sqrt{n}$ radial profile. Hence every vanishing of an accumulation is a fiber/pairing cancellation (orthogonality, Proposition 12.8), never a collapse of the carrier.

Theorem 7.4 (Per-height channels; `finiteCarrier_critical_tendsto`, `eta_critical_tendsto`, `perHeight_channel_nonprincipal`, `perHeight_channel_zeta`). For every ordinate $y \in \mathbb{R}$:

- (i) for non-principal χ the raw weighted partial sums $\sum_{n < N} \chi(n) n^{-(1/2+iy)}$ converge to $L(\frac{1}{2} + iy, \chi)$ (`finiteCarrier_critical_tendsto`), so the convergent channel's nonvanishing factor is 1;
- (ii) for the principal character (and ζ) the raw series diverges, but the alternating (eta) channel $\sum_{n < N} (-1)^{n+1} n^{-s}$ converges to $(1 - 2^{1-s}) \zeta(s)$, whose correction factor is nonzero on the line (`eta_critical_tendsto`, `zeta_eta_factor_ne_zero`).

In both cases the channel value vanishes exactly when the L -value does (`LFunction_critical_eq_zero_iff_pairing_riemannZeta_critical_eq_zero_iff_eta_tendsto_zero`), the on-line limit $y \mapsto L(\frac{1}{2} + iy, \chi)$ is continuous (`continuous_LOnLine`), and it is the analytic limit of the finite carrier (`finiteCarrier_tendsto_LOnLine`).

This is the precise, defensible form of “every zero on the line is subject to the cancellation machine”: it asserts nothing about *where* the zeros are, only that each on-line zero is exactly a fiber cancellation of the convergent channel.

7.2 A fully convergent worked fibre for ζ

For the Riemann zeta channel the convergence above is realized as an explicit, quantitatively controlled Cauchy sequence. Place the fibre at the critical-line point $\text{sFiber}(T)$, which has real part $\frac{1}{2}$ by construction (`sFiber_re`).

Theorem 7.5 (The η -fibre converges on the line; `Pi3Fiber.PartialFiber_tendsto_FullFiber`, `PartialFiber_cauchySeq`, `FullFiber_eq_zero_iff_zeta`). *The alternating partial sums $\text{PartialFiber}(N, T) = \sum_{n < N} (-1)^n (n+1)^{-\text{sFiber}(T)}$ form a Cauchy sequence (`PartialFiber_cauchySeq`) converging to $\text{FullFiber}(T) = \eta(\text{sFiber}(T))$ (`PartialFiber_tendsto_FullFiber`, with vanishing tail `TailFiber_tendsto_zero`); and it vanishes iff $\zeta(\text{sFiber}(T)) = 0$ (`FullFiber_eq_zero_iff_zeta`). The phasor form of the partial sum exhibits the spin $\exp(-(HT \log(n+1))i)$ on the amplitude $(n+1)^{-1/2}$ (`PartialFiber_eq_phasor`).*

Proof sketch. The summand differences are controlled by the consecutive-power bound $\|a^{-w} - (a+1)^{-w}\| \leq \|w\| a^{-\text{Re } w - 1}$ for $a \geq 1$, $\text{Re } w > 0$ (`cpow_neg_consecutive_norm_le`), giving summability of the bracketed series (`pairedTerm_summable`) and the Cauchy property on $\text{Re } s > 0$. The bracketed sum agrees with η for $\text{Re } s > 1$ and extends by the identity theorem across the preconnected region $\{\text{Re } s > 0\} \setminus \{1\}$ (`pairedEta_eq_etaTrivial`, `rightHalfPlane_sdiff_one_isPreconnected`, `sFiber_ne_one`). $\square$

Remark 7.6 (Exact accumulation, not approximate evaluation). *The critical-line behavior of L is classically reached by approximate means — the approximate functional equation and the Riemann–Siegel formula — which represent $L(\frac{1}{2} + it)$ by truncated sums with a controlled error term. The accumulation here is exact: the regularized carrier converges to L on the line (Theorems 7.4, 7.5) and vanishes iff L does (Theorem 9.5), an identity with no error term, machine-checked. Correspondingly the vanishing is an exact event — the sampled eigenwave is exactly ℓ^2 -orthogonal to the arithmetic data (Proposition 12.8) — rather than an approximate minimum of $|L|$. The exactness is what lets the carrier’s geometric and spectral structure transfer to L rigorously rather than heuristically; it is the foundation the rest of the development stands on. (The underlying convergence is the classical Abel/ η summation on $\text{Re } s > 0$; what is new is the exact, machine-checked geometric carrier built on it.)*

7.3 Python carrier-fiber method for locating vanishings

The finite Python harness `carrier_fiber/carrier_fiber.py` separates the carrier geometry from the analytic calibration. Its native finite path uses the pitch-one $\pi/3$ carrier directly: a height $z > 0$ absorbs

$$N = \lfloor z \rfloor$$

complete integer slots and the fractional part $\alpha = z - N$ of the next slot. With a real periodic coefficient $c(n)$, the signed finite fiber term is

$$v_n = c(n) ((\pi/3)n)^{-1/2} \exp(i(n-1)\pi/3).$$

The whole-fiber focal value at height z is therefore the piecewise-linear path

$$B(z) = \sum_{n=1}^N v_n + \alpha v_{N+1}.$$

The readout ordinate printed by the harness is $\log z$; the height itself is the carrier coordinate.

The native scan `find_continuous_focal_crossings` finds candidate vanishings without a one-dimensional nonlinear root solver. On each interval $[N, N + 1]$ it writes $S_N = \sum_{n=1}^N v_n$ and minimizes the distance from the origin to the segment $S_N + \alpha v_{N+1}$ by the explicit projection

$$\alpha_N = \text{clip}_{[0,1]} \left(-\frac{\text{Re}(S_N \overline{v_{N+1}})}{|v_{N+1}|^2} \right).$$

The residual is

$$r_N = |S_N + \alpha_N v_{N+1}|.$$

Heights with r_N below the requested tolerance are reported as exact finite focal crossings; if no exact crossing is present in the scan range, the harness reports the smallest residuals. In compact form:

```
terms = c(n) * ((pi/3) * n)**(-0.5) * exp(1j * slot * pi/3)
prefix[N] = sum_{n <= N} terms[n]
alpha[N] = clip(-real(prefix[N] * conj(terms[N+1])) / abs(terms[N+1])**2, 0, 1)
height[N] = N + alpha[N]
residual[N] = abs(prefix[N] + alpha[N] * terms[N+1])
```

The same scan also evaluates the harmonic pencil marker. The unsigned admissibility channel is

$$A(z) = \sum_{n=1}^N |c(n)| ((\pi/3)n)^{-1/2} + \alpha |c(N+1)| ((\pi/3)(N+1))^{-1/2}.$$

For each pencil

$$H(z) = \begin{pmatrix} A(z) & B(z) \\ \mu A(z) & \lambda B(z) \end{pmatrix},$$

the code checks

$$\det H = (\lambda - \mu)A(z)B(z), \quad \det(H^*H) = |\det H|^2.$$

Thus, whenever $A(z) \neq 0$ and $\lambda \neq \mu$, the Gram rank-drop marker is exactly the same event as the finite focal cancellation $B(z) = 0$.

The default validation command uses the cumulative `c1` carrier clock, which is the Lean-aligned readout used for the L -channel:

$$T = \gamma/(\pi/3), \quad B_N(T) = \frac{\pi}{3} \sum_{n \leq N} c(n) n^{-1/2} \exp(-i(\pi/3)T \log n).$$

In finite mode the harness evaluates this finite sum and locally minimizes $|B_N(T)|^2$ by a coarse grid followed by golden-section refinement in the window around $T = \gamma/(\pi/3)$. In analytic mode it uses the exact calibrated channels

$$A(T) = L(3/2 + i\gamma, \chi), \quad B(T) = (\pi/3)L(1/2 + i\gamma, \chi),$$

then applies the same determinant and Gram determinant checks. The reference γ table selects which displayed ordinates to validate; the focal marker remains $B = 0$ and the pencil marker remains $\det H = 0$.

8 The geometric explicit formula

The logarithmic derivative of the fiber is the prime-power accumulation weighted by the winding.

Theorem 8.1 (Prime field; `explicit_formula_prime_field`, `dirichlet_logDeriv_eq_tsum`). For $\text{Re } s > 1$,

$$-\frac{L'(s, \chi)}{L(s, \chi)} = \sum_n \chi(n) \Lambda(n) n^{-s},$$

where Λ is the von Mangoldt function. The zeros of $L(\cdot, \chi)$ are the poles of the meromorphic continuation of this object.

Definition 8.2 (Residue harmonic; `Residue.residueHarmonic`). For $x > 0$ and $\gamma \in \mathbb{R}$, write $\rho = \frac{1}{2} + i\gamma$ and set $\text{residueHarmonic}(x, \gamma) = -x^\rho/\rho$.

Theorem 8.3 (Zeros located as poles; `zeta_zero_located_as_pole`, `L_zero_located_as_pole`). Let $\rho = \frac{1}{2} + i\gamma$ be a zero of $L(\cdot, \chi)$ at which $L'(\rho, \chi) \neq 0$ (a simple zero). Then ρ is a simple pole of the explicit-formula kernel $-(L'/L)(s) x^s/s$, with

$$\lim_{s \rightarrow \rho} (s - \rho) \left(-\frac{L'(s, \chi)}{L(s, \chi)} \cdot \frac{x^s}{s} \right) = \text{residueHarmonic}(x, \gamma) = -\frac{x^\rho}{\rho}.$$

The same statement holds verbatim for ζ .

Proof sketch. On the punctured neighborhood of ρ in $\{s \neq 1\}$, L is analytic with $L(\rho) = 0$ and $L'(\rho) \neq 0$, so $-(L'/L)$ has a simple pole at ρ with residue 1; multiplying by the analytic factor x^s/s scales the residue to x^ρ/ρ and the leading sign gives $-x^\rho/\rho$. $\square$

9 Projection and reality on the critical line

The carrier is three-dimensional; the readout projects it to a one-dimensional coordinate. We first record that coordinate and the change of scale it carries (Section 9.1), then the reality of the completed object that the readout returns (Section 9.2).

9.1 The readout coordinate and its scaling

The carrier is built in its native gauge: the unit is $\pi/3$, and a full angular turn is six such units, $6 \cdot (\pi/3) = 2\pi$ (`spin_phasor_period6`, Theorem 3.13). The readout that sends a three-dimensional harmonic value $t \in \mathbb{R}$ down to a one-dimensional coordinate composes a Cayley map [9] with a rescaling to the standard unit interval (`HarmonicProjection`):

$$\text{toCircleAngle}(t) = 2 \arctan t \in (-\pi, \pi), \quad \text{toLine}(\theta) = \frac{\theta + \pi}{2\pi} \in (0, 1),$$

and $\text{projection} = \text{toLine} \circ \text{toCircleAngle}$. The second map carries the change of scale: it sends the full turn 2π to length 1, so the six $\pi/3$ buckets of the gauge become six equal sub-intervals of the readout. Under this rescaling the coordinate is rigid — the Cayley map is odd (`toCircleAngle_odd`), the normalization is affine hence midpoint-preserving (`toLine_midpoint`), and the principal-arc ends $\theta = \pm\pi$ map to the two ends of the interval (`toLine_neg_pi`, `toLine_pi`) — so the centre of the gauge is carried to the centre of the interval.

Theorem 9.1 (Coordinate centre; `HarmonicProjection.projection_midline`). *projection(0) is the midpoint of the readout interval: the centre of the six gauge buckets, the angle $\theta = 0$, maps to the average of the two endpoint images (`toLine_midpoint_of_endpoints`).*

The inverse conversion is the same scaling read backwards: a readout coordinate u is the angle $\theta = 2\pi u - \pi$, the gauge reading $6u - 3$ in $\pi/3$ units, and the interval's centre is the gauge centre $\theta = 0$. This is a property of the readout coordinate and its scaling. The value the readout returns on the completed object is real — it lands on the real axis (Section 9.2).

9.2 Reality on the line

The supporting analytic facts are the conjugation (Schwarz reflection [9]) and functional symmetries of Λ . The $\frac{1}{2}$ in this subsection is the *reflection axis* $\operatorname{Re} s = \frac{1}{2}$ — the fixed line of $s \mapsto 1 - s$ on which $\bar{s} = 1 - s$ (Section 2) — *not* the scale-critical exponent of Section 3.3: the reality is read off the functional equation, not the area law.

Theorem 9.2 (Conjugation symmetry; `HelixCollapse.completedRiemannZeta_conj`). *For all $s \in \mathbb{C}$, $\overline{\Lambda(\bar{s})} = \Lambda(s)$.*

Proof sketch. On $\operatorname{Re} s > 1$ this is the real Dirichlet coefficients of Λ ($\overline{\Lambda(\bar{s})} = \Lambda(s)$ termwise, using $\overline{\pi^{-s/2}} = \pi^{-\bar{s}/2}$ and $\overline{\Gamma(\bar{s}/2)} = \Gamma(s/2)$); the identity extends to $\mathbb{C}$ by analytic continuation of Λ_0 (the pole-removed completion) via the identity theorem. $\square$

Theorem 9.3 (Reality on the line; `completedRiemannZeta_critical_line_im_zero`). *For every $t \in \mathbb{R}$, $\operatorname{Im} \Lambda(\frac{1}{2} + it) = 0$.*

Proof sketch. On the line $\overline{\frac{1}{2} + it} = 1 - (\frac{1}{2} + it)$, so Theorem 9.2 together with $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) gives $\overline{\Lambda(\frac{1}{2} + it)} = \Lambda(\frac{1}{2} + it)$, i.e. a real value. $\square$

Theorem 9.4 (Faithful projection for ζ ; `faithful_projection_zeta`). *For every $\gamma \in \mathbb{R}$,*

$$\operatorname{Im} \Lambda(\tfrac{1}{2} + i\gamma) = 0 \quad \text{and} \quad F_\eta(\gamma) = 0 \iff \zeta(\tfrac{1}{2} + i\gamma) = 0,$$

where $F_\eta(\gamma) = (1 - 2^{1-s})\zeta(s)$ at $s = \frac{1}{2} + i\gamma$ (`EtaTrivial.Feta`).

Theorem 9.5 (Located crossings lie on the line; `crossing_is_zero_on_line`). *If the eta-regularized limit of Theorem 7.3 vanishes at ordinate y , then $\zeta(\frac{1}{2} + iy) = 0$ and $\operatorname{Re}(\frac{1}{2} + iy) = \frac{1}{2}$.*

10 Faithfulness for every Dirichlet L -function and Tate completion

A single carrier serves all χ ; the character weight $\chi(n)$ on each phasor is the only dial.

Theorem 10.1 (One carrier, all characters; `faithful_all_L`). *For every Dirichlet character χ :*

- (i) *for $\operatorname{Re} s > 1$, $\sum_{n < N} \chi(n) n^{-s} \rightarrow L(s, \chi)$; and*
- (ii) *on the critical line $\operatorname{Re} s = \frac{1}{2}$, the eta-twisted closed carrier `etaTwistClosed`(χ, s) vanishes if and only if $L(s, \chi) = 0$ (`etaTwistClosed_eq_zero_iff_critical`).*

Definition 10.2 (Completed carrier; `Tate.completedCarrier`). *For a Dirichlet character χ , set $\Lambda_\chi^c(y) = \Lambda(\chi, \frac{1}{2} + iy)$ using the archimedean Γ -completion $\Lambda(\chi, s) = (\text{gamma factor}) \cdot L(s, \chi)$.*

Theorem 10.3 (Completion and functional equation; `faithful_all_L_completed`, `completedCarrier_eq_zero_i`, `completed_functional_equation`). For every primitive χ modulo q :

- (i) the completion is zero-free off the zeros of L on the line: $\Lambda_\chi^c(y) = 0 \iff L(\frac{1}{2} + iy, \chi) = 0$ (the Γ -factor never vanishes for $\operatorname{Re} s > 0$); and
- (ii) Tate's functional equation [12] holds: $\Lambda(\chi, 1 - s) = q^{s-1/2} W(\chi) \Lambda(\chi^{-1}, s)$, where $W(\chi) = \operatorname{rootNumber}(\chi)$.

11 The double-ended carrier and conjugacy

The carrier is chiral: the right strand winds with spin $e^{-iy \log n}$, and its mirror, the left strand, winds with $e^{+iy \log n}$.

Theorem 11.1 (Conjugate chiralities; `both_helices_conjugate`). For every $N \in \mathbb{N}$ and $y \in \mathbb{R}$,

$$\sum_{n < N} (-1)^{n+1} n^{-(1/2-iy)} = \overline{\sum_{n < N} (-1)^{n+1} n^{-(1/2+iy)}}.$$

Proof sketch. Termwise: $\overline{(-1)^{n+1}} = (-1)^{n+1}$ and $\overline{n^{-(1/2+iy)}} = n^{-(1/2-iy)}$ for the positive real base n (so $\arg n \neq \pi$), by `Complex.cpow_conj` and $\bar{n} = n$. $\square$

Corollary 11.2 (Equal modulus, identical crossing heights). The two strands have equal modulus at every N , so their limits vanish at the same ordinates y .

Theorem 11.3 (Completed sides agree; `both_sides_cross_together`). For every $y \in \mathbb{R}$, $\Lambda(\frac{1}{2} - iy) = \Lambda(\frac{1}{2} + iy)$.

Proof sketch. $\frac{1}{2} - iy = 1 - (\frac{1}{2} + iy)$, so this is $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) on the line. $\square$

12 The Frobenius similitude and the determinant identity

The map $n \mapsto pn$ — and more generally $n \mapsto mn$ — is the *Frobenius self-similarity* of the carrier. We define it precisely, as a similitude of the carrier line.

Definition 12.1 (Frobenius similitude; `frobeniusMultiplier`, `frobeniusSimilitude`). For an integer m the Frobenius multiplier is $\mu_\theta(m) = \sqrt{m} \operatorname{wind}_\theta(m) \in \mathbb{C}$, and the Frobenius similitude $\operatorname{Sim}_\theta(m)$ is the $\mathbb{C}$ -linear self-map $w \mapsto \mu_\theta(m)w$ of the carrier line $\mathbb{C}$.

Theorem 12.2 (Prime multiplication acts by similitude; `helixPt_mul`, `helixPt_eq_similitude`, `norm_frobeniusMultiplier`, `frobeniusSimilitude_mul`). For $m, n \neq 0$, multiplication $n \mapsto mn$ acts on the carrier point $\operatorname{helixPt}_\theta(n) = \sqrt{n} \operatorname{wind}_\theta(n)$ by the similitude

$$\operatorname{helixPt}_\theta(mn) = \operatorname{Sim}_\theta(m)(\operatorname{helixPt}_\theta(n)) = \underbrace{\sqrt{m}}_{\text{radial}} \underbrace{\operatorname{wind}_\theta(m)}_{\text{rotation}} \operatorname{helixPt}_\theta(n).$$

Its multiplier has modulus $\|\mu_\theta(m)\| = \sqrt{m}$ — the area-law factor of Theorem 3.9, the winding rotation being unit-modulus — so the map scales every length by $\sqrt{m}$. Moreover the similitudes compose, $\operatorname{Sim}_\theta(mn) = \operatorname{Sim}_\theta(m) \circ \operatorname{Sim}_\theta(n)$: the carrier is self-similar under the whole multiplicative monoid generated by the primes (the Frobenius maps).

Under the bridge (Theorem 4.4), with prime-angle assignment $\theta(p) = -y \log p$ the rotation eigenphase at ordinate y is the spin $z = \text{spin}(y, n) = e^{-iy \log n} \in \mathbb{T}$ (`frobeniusMultiplier_bridge`). The two chiralities (Theorem 11.1) carry z and $\bar{z}$.

Theorem 12.3 (Conjugate-eigenstate determinant; `frobenius_conjugate_det_one`). *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, with $z = \text{spin}(y, n) = e^{-iy \log n}$,*

$$\det \begin{pmatrix} z & 0 \\ 0 & \bar{z} \end{pmatrix} = z \bar{z} = |z|^2 = 1.$$

Proof. The 2×2 determinant is $z \cdot \bar{z} - 0 \cdot 0 = z \bar{z}$. Since $\bar{z} = \overline{e^{-iy \log n}} = e^{+iy \log n} = e^{+iy \log n}$ and $-iy \log n + \overline{-iy \log n} = 0$, we have $z \bar{z} = e^{-iy \log n} e^{+iy \log n} = e^0 = 1$. $\square$

Theorem 12.4 (Unitary transverse block; `frobeniusBlock_det_one`, `frobeniusBlock_unitary`). *The transverse block $\text{diag}(z, \bar{z})$ assembled from the two chiral eigenphases at a conjugate crossing is unitary, $M^H M = I$, with determinant $z \bar{z} = |z|^2 = 1$ — the orientation- and volume-preserving chiral invariant of the transverse phase dynamics. The radial scaling $\sqrt{m}$ of the similitude (Theorem 12.2) is carried separately and is not part of this block.*

12.1 The self-adjoint eigenstate

The determinant identity above is stated for the bare eigenphases $z, \bar{z} \in \mathbb{T}$. We realize them as the values of a genuine eigenstate of a self-adjoint generator. This is the operator *design*: it is real because its inputs are real, and it stands independently of any zero. Reading the nontrivial zeros as the spectrum of a self-adjoint operator is the *Hilbert–Pólya* program — supported by the random-matrix law of the zero statistics [13] and pursued through the Berry–Keating $H = xp$ quantization [14] and Connes’s noncommutative-geometry trace formula [15]; the operator design here is unconditional and per-height, and makes no claim to be that canonical operator (Remark 12.11).

Definition 12.5 (Spectral wave; `spectralWave`). *For $\gamma \in \mathbb{R}$, $\psi_\gamma(t) = e^{i\gamma t}$ ($t \in \mathbb{R}$).*

Theorem 12.6 (Unit-modulus eigenstate of $D = -i d/dt$; `spectralWave_eigen`, `spectralWave_norm`). *For all $t \in \mathbb{R}$, $-i \psi'_\gamma(t) = \gamma \psi_\gamma(t)$ and $\|\psi_\gamma(t)\| = 1$. Thus ψ_γ is a unit-modulus eigenstate of the generator $D = -i d/dt$ with the real eigenvalue γ .*

The reality is structural, not an accident of ψ_γ . Multiplication by the real height, $\text{vonNeumannOp}(\gamma) = \gamma \cdot \text{id}$ on the fiber $\mathbb{C}$, is symmetric with eigenvalue γ (`vonNeumannOp_isSymmetric`, `vonNeumannOp_hasEigenvalue`); on the finitely-supported sequence space the real-diagonal operator $\text{diagOp}(d)$ is symmetric (`diagOp_symmetric`), with the basis vectors as explicit eigenvectors carrying the real eigenvalues d_i (`diagOp_eigenvector`), and *every* eigenvalue is real (`diagOp_spectrum_real`) by von Neumann reality — a symmetric operator on a complex inner-product space has real eigenvalues (`symmetric_eigenvalue_real`). These are collected in `unconditional_frobenius_eigenstate` and `frobenius_spectralWave_eigenstate`. The eigenwave is a one-parameter unitary group, $\psi_\gamma(s+t) = \psi_\gamma(s) \psi_\gamma(t)$ with $\psi_\gamma(0) = 1$ (`spectralWave_add`, `spectralWave_zero`): the unitary flow generated by the real spectral parameter γ (Stone form).

Proposition 12.7 (The chiralities are eigenstate values; `spin_eq_spectralWave`, `conj_spin_eq_spectralWave`). *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, $\text{spin}(y, n) = \psi_y(-\log n)$ and $\overline{\text{spin}(y, n)} = \psi_y(\log n)$.*

The useful spectral content is not the mere existence of the eigenwave ψ_y — every real y gives one — but its *orthogonality* to the arithmetic data at a vanishing.

Proposition 12.8 (The accumulation is an eigenstate–data inner product; `fiberCarrierFinite_eq_inner`, `fiberCarrierFinite_eq_zero_iff_orthogonal`). Split the critical-line phasor $\chi(n)n^{-(1/2+iy)}$ into the arithmetic datum $\chi(n)n^{-1/2}$ (the finitely-supported vector `dataVec`(χ, N)) and the eigenwave sample $\text{spin}(y, n) = \psi_y(\log n)$ (the vector `waveVec`(y, N)). With the ℓ^2 inner product $\langle f, g \rangle = \sum_n \overline{f(n)} g(n)$ on the finitely-supported sequence space, the finite accumulation is their pairing,

$$\sum_{n < N} \chi(n) n^{-(1/2+iy)} = \langle \text{waveVec}(y, N), \text{dataVec}(\chi, N) \rangle,$$

so it vanishes iff the sampled eigenwave is ℓ^2 -orthogonal to the arithmetic data (`fiberCarrierFinite_eq_zero_iff_orthogonal`). A vanishing point is exactly a real-frequency orthogonality event, exhibited at each N (and the pairing is the partial phasor sum $G_{\chi, N}(\frac{1}{2} + iy)$, which accumulates to the carrier value $L(\frac{1}{2} + iy, \chi)$; `inner_eq_finiteCarrier`). The Gram form of any finite family of such vectors is positive-semidefinite (`carrier_gram_nonneg`), the positive-kernel background for the de Branges evaluation of Section 15.

Theorem 12.9 (Determinant on the eigenstate values; `frobenius_spectralWave_det_one`). For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, $\det \text{diag}(\psi_y(-\log n), \psi_y(\log n)) = 1$.

Theorem 12.10 (Frobenius unimodularity of the produced eigenstate; `frobenius_eigenstate_det_one`). Fix $\gamma \in \mathbb{R}$, $n \in \mathbb{N}$, and $\psi : \mathbb{R} \rightarrow \mathbb{C}$, and assume the vanishing $\Rightarrow$ eigenstate link $\psi = \psi_\gamma$. Then ψ is a unit-modulus eigenstate of $D = -i d/dt$ with eigenvalue γ , and $\det \text{diag}(\psi(-\log n), \psi(\log n)) = 1$.

Remark 12.11 (The open link). We exhibit the eigenstate ψ_γ for every real γ unconditionally (Theorem 12.6). What is not proved is that a carrier vanishing at height γ produces this eigenstate; that hypothesis ($\psi = \psi_\gamma$) is the one input of Theorem 12.10, left open. We do not derive that the vanishings are the spectrum of a canonical operator.

13 The chiral cup and its Hilbert completion

The pairing underlying the on-line spectral reading is the *chiral cup*, a Hermitian form on the finitely-supported fibre space $\ell_0 = (\mathbb{N} \rightarrow_0 \mathbb{C})$ assembled from the forward and reverse (right/left) chiralities of the double-ended carrier (Section 11).

Definition 13.1 (Chiral defect and cup; `CupIdentity.Dop`, `Cup`, `Jconj`, `iotaR`, `iotaL`). With the right/left site embeddings ι_R, ι_L and the conjugation J , the chiral defect is $\text{Dop } F = \iota_R F - J(\iota_L F)$ — the failure of the forward embedding to match the conjugated reverse — and the cup is the sesquilinear pairing $\text{Cup}(F, G) = \langle \text{Dop } F, \text{Dop } G \rangle$.

Theorem 13.2 (The cup is a definite Hermitian form; `cup_hermitian`, `cup_positive_semidefinite`, `cup_self_im`, `cup_nullspace_safe`, `cup_null_iff`). Cup is Hermitian ($\text{Cup}(G, F) = \overline{\text{Cup}(F, G)}$), positive semidefinite ($0 \leq \text{Re } \text{Cup}(F, F)$), real on the diagonal ($\text{Im } \text{Cup}(F, F) = 0$), and definite: $\text{Cup}(F, F) = 0 \iff \text{Dop } F = 0 \iff F = 0$ (`cup_null_iff`, `dop_eq_zero_iff_carrier_zero`). The closure therefore preserves all arithmetic content (`closure_preserves_arithmetic_content`): it never collapses a nonzero fibre, and the readout Dop is injective (`spectral_readout_faithful`).

Theorem 13.3 (Hilbert completion; `hilbert_completion_exists`, `cup_self_eq_two_energy`). The cup completes to a genuine Hilbert space: there is a complete inner-product space into which Dop embeds with $\langle \text{Dop } F, \text{Dop } G \rangle = \text{Cup}(F, G)$. On the diagonal the cup is the ℓ^2 energy, $\text{Cup}(F, F) = 2 \sum_n \|F(n)\|^2$.

Theorem 13.4 (Frobenius transport is a cup similitude; `Tw`, `transport_cup_modulus`, `frobenius_weighted_cup_identity`, `transport_unitary_after_normalization`). *The weight transport T_μ scales the cup by $\|\mu\|^2$, $\langle T_\mu(\text{Dop } F), T_\mu(\text{Dop } G) \rangle = \|\mu\|^2 \text{Cup}(F, G)$, and a unit weight ($\|\mu\| = 1$) transports unitarily (`transport_unitary_after_normalization`). With $\mu = \text{amp} \cdot \text{sgn}$, $\|\text{sgn}\| = 1$, the cup scales by amp^2 : this is the cup-level form of the Frobenius similitude (Theorem 12.2), whose radial factor $\sqrt{m}$ squares to the area-law weight m .*

Theorem 13.5 (Von Neumann reality and modulus balance; `von_neumann_reality`, `diagOp_spectrum_real`, `midpoint_forcing`). *Every eigenvalue of a symmetric operator on a complex inner-product space is real (`von_neumann_reality`), and the real-diagonal operator `diagOpd` has real spectrum (`diagOp_spectrum_real`). The cup-level reflection pins the readout to the line: for $q > 1$, if $\|q^s\|^2 = q$ then $\text{Re } s = \frac{1}{2}$ (`midpoint_forcing`) — since $\|q^s\|^2 = q^{2\text{Re } s} = q$ forces $2\text{Re } s = 1$, a balance-to-line implication independent of any zero.*

Theorem 13.6 (No chiral dominance, by crossing induction; `no_dominance_by_crossing_induction`, `crossing_balance_preserved`, `crossing_no_dominance_cup`). *For a crossing helix with a positive Frobenius cup, weighted forward/reverse transport, and zero-readout exhaustion, the modulus balance $\|\text{fwd}(n)\| = \|\text{rev}(n)\|$ propagates from each crossing to the next (`crossing_balance_preserved`); hence no chirality dominates at any crossing (`no_dominance_by_crossing_induction`) and the forward and reverse cup energies agree at every site (`crossing_no_dominance_cup`). The hypotheses are inhabited and non-vacuous: a balanced helix satisfies all of them, while an unbalanced one fails exhaustion (`balancedHelix_exhaust`, `unbalancedHelix_not_exhaust`).*

Remark 13.7 (The cup carries the von Mangoldt prime field; `PrimePowerProbe.cup_intertwining`, `vonMangoldtChannel_eq_neg_logDeriv`). *On the prime-power probe fibre the cup energy is exactly the squared norm of the winding-weighted von Mangoldt terms, $\text{Cup}(\text{probe}, \text{probe}) = 2 \sum_{n < N} \|\text{channelTerm}(n)\|^2$ (`cup_intertwining`), supported on prime powers (`channelTerm_eq_zero_of_not_isPrime`) whose superposition is $-L'/L$ for $\text{Re } s > 1$ (`vonMangoldtChannel_eq_neg_logDeriv`); in that region the channel never vanishes (`noDominance_no_offline_zero`). This is the explicit formula of Section 8 read through the cup energy.*

14 The Kähler polarization, the trace formula, and a cohomological re-expression of the zeros

The fibre carries a compatible triple — a symplectic form, a complex structure, and a metric — under which the Frobenius maps act as conformal symplectic similitudes, the von Mangoldt prime field appears as an operator trace, and the vanishing of L is re-expressed as a one-dimensional cohomology.

14.1 The polarization and its Kähler identity

Definition 14.1 (Symplectic form, complex structure, metric; `HelixPolarization.OmegaH`, `JH`, `gH`). *On the fibre $H = (\mathbb{N} \rightarrow_0 \mathbb{C})$: the real antisymmetric $\mathbb{R}$ -bilinear form Ω (`OmegaH_antisymm`, `OmegaH_add_right`, `OmegaH_smul_right`), the almost-complex structure J with $J^2 = -\text{id}$ (`JH_sq`), and the metric g .*

Theorem 14.2 (Kähler compatibility; `OmegaH_JH`, `gH_symm`, `OmegaH_compat`, `inner_eq_gH_add_I_OmegaH`). *The triple is compatible: $\Omega(x, Jy) = g(x, y)$, g is symmetric, and J preserves Ω ($\Omega(Jx, Jy) =$*

$\Omega(x, y)$). The Hermitian inner product splits into its metric and symplectic parts,

$$\langle x, y \rangle = g(x, y) + i \Omega(x, y),$$

the Kähler identity (*inner_eq_gH_add_I_OmegaH*).

Theorem 14.3 (Positivity of the polarization; *theoremA*, *gH_Dop_self*). For $F \neq 0$, $0 < \Omega(\text{Dop } F, J \text{Dop } F)$: the polarization is strictly positive on the chiral defect (*theoremA*). The metric recovers the cup energy, $g(\text{Dop } F, \text{Dop } F) = \text{Re Cup}(F, F)$ (*gH_Dop_self*).

14.2 Frobenius as a conformal symplectic similitude

Theorem 14.4 (Frobenius scales the symplectic form by m ; *frobT_weighted_adjoint*, *theoremB*, *frobT_mul*, *siteShift_weighted_adjoint*, *siteShift_mul*). The Frobenius transport $\text{frobT}(m, \text{sgn})$ (unit sign) scales the inner product by m , $\langle \text{frobT } x, \text{frobT } y \rangle = m \langle x, y \rangle$ (*frobT_weighted_adjoint*), and on the polarization it scales the symplectic pairing by exactly p^r ,

$$\Omega(\text{frobT}(p^r) x, J \text{frobT}(p^r) y) = p^r \Omega(x, Jy) \quad (\text{theoremB}).$$

The maps compose multiplicatively, $\text{frobT}(mn, s_1 s_2) = \text{frobT}(m, s_1) \circ \text{frobT}(n, s_2)$ (*frobT_mul*). The companion geometric site shift $\text{siteShift}(m) : \delta_n \mapsto \sqrt{m} \delta_{mn}$ is the same $\sqrt{m}$ similitude on sites, unitary up to the m -factor and multiplicative (*siteShift_weighted_adjoint*, *siteShift_mul*) — the operator avatar of $\text{helixPt}(mn) = \sqrt{m} \text{wind}(m) \text{helixPt}(n)$.

14.3 The von Mangoldt trace

Theorem 14.5 (The weighted trace is $-L'/L$; *theoremC*, *traceWeighted_eq_superposition*, *theoremC_trace*, *specDiag_siteShift*, *localTrace_mul*). The weighted local trace is the prime-power superposition $\text{traceWeighted}(\sigma, h) = \sum_n \chi(n) \Lambda(n) \langle \delta_n, \text{specDiag } \delta_n \rangle$ (*theoremC_trace*), equal to the von Mangoldt channel (*traceWeighted_eq_superposition*), which for $\sigma > 1$ is the logarithmic derivative $-L'(\cdot)/L(\cdot)$ at the spectral parameter (*theoremC*). The local factor is completely multiplicative (*localTrace_mul*) and intertwines with the site shift, $\text{specDiag} \circ \text{siteShift}(m) = \text{localTrace}(m) \cdot (\text{siteShift}(m) \circ \text{specDiag})$ (*specDiag_siteShift*). This is the geometric explicit formula (Section 8) realized as an operator trace.

14.4 A cohomological re-expression of the vanishing

Theorem 14.6 (L vanishes iff the helix cohomology is nontrivial; *theoremD*, *theoremD_H0_finrank*, *theoremD_H1_finrank*, *theoremD_diff*). The helix cohomology at ρ is the quotient $H^\bullet(\rho) = \mathbb{C}/\langle L(\rho, \chi) \rangle$, with the eigenclass the class of 1. Then

$$L(\rho, \chi) = 0 \iff \text{eigenclass}(\rho) \neq 0 \iff H^1(\rho) \text{ nontrivial},$$

and the cohomology dimension is the exact indicator $\dim_{\mathbb{C}} H^0(\rho) = \dim_{\mathbb{C}} H^1(\rho) = \mathbf{1}[L(\rho, \chi) = 0]$ (*theoremD_H0_finrank*, *theoremD_H1_finrank*): a simple zero is a one-dimensional class, a non-zero is acyclic.

Remark 14.7 (The cohomology is the quotient $\mathbb{C}/\langle L \rangle$). Concretely $H^\bullet(\rho) = \mathbb{C}/\langle L(\rho, \chi) \rangle$, so Theorem 14.6 is the field-theoretic identity “ $\mathbb{C}/\langle x \rangle \neq 0 \iff x = 0$ ” at $x = L(\rho, \chi)$ — a cohomological re-expression of the vanishing. The analytic weight of this section is carried by the polarization (Theorem 14.2), the Frobenius scaling (Theorem 14.4), and the trace identity (Theorem 14.5).

Theorem 14.8 (Germ cohomology dimension; `germH1_finrank`, `jetRead_surjective`, `ker_jetRead_eq_range_germCoboundary`). *The n -jet germ cohomology has dimension exactly n : $\dim_{\mathbb{C}} H_{\text{germ}}^1(n) = n$ (`germH1_finrank`), via the surjective jet readout (`jetRead_surjective`) and $\ker(\text{jetRead}) = \text{range}(\text{germCoboundary})$ (`ker_jetRead_eq_range_germCoboundary`): the local jet data of order n is an n -dimensional obstruction space.*

15 The de Branges evaluation

The reality on the critical line (Section 9.2) and the eigenstate above sit inside the Hermite–Biehler / de Branges spectral framework [10, 11]. We record the framework, its reality and discreteness consequences, a worked example, and the change of variables that places the on-line cancellations as real spectral points.

Definition 15.1 (Structure function and components; `Estar`, `Acomp`, `Bcomp`, `IsHB`). *For $E : \mathbb{C} \rightarrow \mathbb{C}$, the reflection is $E^*(z) = \overline{E(\bar{z})}$, the components are $A = \frac{1}{2}(E + E^*)$ and $B = \frac{i}{2}(E - E^*)$, and E is Hermite–Biehler (`IsHB E`) when $\|E^*(z)\| < \|E(z)\|$ for every z with $\text{Im } z > 0$.*

Proposition 15.2 (Gram positivity; `gram_quadratic_form_nonneg`). *For vectors v_i in an inner-product space and scalars c_i , $\sum_{i,j} \bar{c}_i c_j \langle v_i, v_j \rangle = \langle \sum_i c_i v_i, \sum_j c_j v_j \rangle \geq 0$ — the positive-definite reproducing-kernel fact behind $H(E)$.*

Theorem 15.3 (A positive feature map forces Hermite–Biehler; `CupIdentity.deBrangesKernel`, `deBrangesKernel_diag`, `featureMap_forces_HB`, `no_featureMap_of_not_HB`). *The de Branges kernel has diagonal $K_E(w, w) = (\|E(w)\|^2 - \|E^*(w)\|^2)/(4\pi \text{Im } w)$ (`deBrangesKernel_diag`). If a feature map φ reproduces it — $\langle \varphi(z), \varphi(w) \rangle = K_E(w, z)$ for all z, w — then on the open upper half-plane $\|E^*(w)\| \leq \|E(w)\|$, i.e. the Hermite–Biehler inequality (`featureMap_forces_HB`), and the kernel diagonal is nonnegative there (`featureMap_forces_diag_nonneg`). Contrapositively, no positive (reproducing-kernel) feature map can exist where HB fails (`no_featureMap_of_not_HB`). Thus HB is precisely the condition for the structure kernel to be a genuine inner-product Gram kernel — the positive-kernel background (Proposition 15.2) made into a dichotomy.*

Theorem 15.4 (Positivity forces a real spectrum; `hb_no_zero_upper`, `Acomp_zero_im_eq_zero`, `Bcomp_zero_im_eq_zero`). *If `IsHB E` then E has no zeros in the open upper half-plane, and every zero of A and of B is real.*

Theorem 15.5 (Discreteness; `Bcomp_zeros_discrete`). *If, in addition, E is entire, then the zeros of B are isolated: a real, discrete spectrum.*

Proposition 15.6 (Paley–Wiener example; `paleyWiener_isHB`, `paleyWiener_Bcomp`). *$E(z) = e^{-iz}$ is Hermite–Biehler, and its B -component is $\sin$, whose spectrum is $\{k\pi\}$.*

Theorem 15.7 (The unconditional Paley–Wiener template; `paleyWiener_no_zero_upper`, `paleyWiener_balance_iff_real`, `paleyWiener_unimodular_block_unifies`, `paleyWiener_Bcomp_zeros_discrete`). *For this genuine Hermite–Biehler E all the conditional readings of the paper hold unconditionally and coincide. There are no zeros in the upper half-plane (`paleyWiener_no_zero_upper`), the structure is rigid off the real axis (`paleyWiener_carrierRigid`), and the three readings are mutually equivalent and equivalent to reality:*

$$\det \text{conjPairBlock}(E(z)/E^*(z)) = 1 \iff \|E(z)\| = \|E^*(z)\| \iff \text{Im } z = 0$$

(*paleyWiener_unimodular_block_unifies*, *paleyWiener_balance_iff_real*, *paleyWiener_conjugate_det_one*). Its B -component zeros are real (*paleyWiener_Bcomp_zero_im_eq_zero*) and discrete (*paleyWiener_Bcomp_zeros_d*) the spectrum $\{k\pi\}$ of $\sin$. This is the unconditional model of exactly what an HB structure function delivers — the unitary conjugate-pair block (Theorem 12.4) and the critical-line readout (Corollary 15.9) become a single equivalence; for Λ the identical package is conditional on $\text{IsHB } \Lambda$, i.e. RH (Remark 15.10).

Theorem 15.8 (Critical-line change of variables; *deBranges_var_im*). For $\rho \in \mathbb{C}$, $\text{Im}(-i(\rho - \frac{1}{2})) = \frac{1}{2} - \text{Re } \rho$; so $z = -i(\rho - \frac{1}{2})$ is real iff $\text{Re } \rho = \frac{1}{2}$.

Corollary 15.9 (On-line cancellations are real spectral points; *criticalLine_deBranges_real*, *zeroHarmonic_deBranges_real*). A carrier vanishing at height γ sits at $\rho = \frac{1}{2} + i\gamma$, whose de Branges variable $z = -i(\rho - \frac{1}{2}) = \gamma$ is real. The same-height cancellations we find are real de Branges spectral points, evaluable in the reality/discreteness framework of Theorems 15.4–15.5.

Remark 15.10 (Scope: $\text{IsHB } \Lambda$ is classical open work, not used here). The generic theorems above are conditional on $\text{IsHB } E$; we never apply them to a Λ -built structure function, and indeed no such structure function is constructed in this work. That $\text{IsHB } \Lambda$ — no zeros off the critical line (*hb_no_zero_upper*) — is equivalent to the Riemann Hypothesis is the classical de Branges reformulation [11]; discharging it is open work in that program, which this paper neither formalizes, assumes, nor rests on. What the paper uses is only the unconditional on-line content — the de Branges variable is real on the critical line (Corollary 15.9), IsHB -free. We make no claim about off-line zeros, and the paper's open question is projection primacy (Remark 17.7), not $\text{IsHB } \Lambda$.

16 The harmonic pencil and the focal eigenheight

The positive/negative channels of the 3D phasor (Section 6) assemble into a 2×2 determinantal pencil whose rank-drop locates the zeros — built in the geometric height $Z = e^y > 0$ rather than in $\log n$ (the *no-log-trap* design). The readout dictionary is $t = \log Z$, and the critical readout $\text{criticalReadout}(Z) = \frac{1}{2} + i \log Z$ has real part $\frac{1}{2}$ for every real Z (*criticalReadout_re*).

16.1 The scalar channel closure

Definition 16.1 (Signed and unsigned harmonic channels; *HarmonicCell.Achan*, *Bchan*, *Focal.Pchan*, *Mchan*). With amplitude $a_n(t) = (\pi/3)n^{-1/2} \exp(-it \log n)$, the positive and negative scalar channels are $P(t) = \sum_{\chi(n)=+1} a_n(t)$ and $M(t) = \sum_{\chi(n)=-1} a_n(t)$. The signed harmonic mode is $B_\chi(Z) = (\pi/3) L(\frac{1}{2} + i \log Z, \chi)$ — the Abel limit of the channel difference $P - M$. The factor $\pi/3$ is the exact six-cell unit of the carrier (*Ucell*); it is nonzero (*Ucell_ne_zero*), so multiplying by it preserves the zero set exactly. The unsigned mode is $A_\chi(Z) = L(\frac{3}{2} + i \log Z, \chi)$, read at the absolute-convergence abscissa $\sigma = \frac{3}{2}$ where it never vanishes (*Achan_ne_zero*).

Theorem 16.2 (Channel closure $\iff$ L -zero; *Focal.channel_diff_tendsto*, *channel_closure_iff_L_zero*). The scalar channel difference is the readout phasor partial sum, with Abel limit $(\pi/3) \Phi_\chi(Z) = (\pi/3) L(\frac{1}{2} + i \log Z, \chi)$ (*channel_diff_tendsto*). Hence the channel closes — the difference $P(\log Z) - M(\log Z)$ tends to 0 — iff $L(\frac{1}{2} + i \log Z, \chi) = 0$ (*channel_closure_iff_L_zero*). The $\pi/3$ scaling is what makes this a literal equality of cancellation events: the harmonic channel is a fixed nonzero multiple of the exact L -fiber, not a numerically tuned proxy.

16.2 The determinantal pencil

Theorem 16.3 (Harmonic pencil determinant; `HarmonicCell.harmonicPencil_det`, `harmonic_pencil_det_zero`). For the diagonal harmonic pencil

$$H_\chi = \begin{pmatrix} A_\chi & B_\chi \\ \mu A_\chi & \lambda B_\chi \end{pmatrix}, \quad \det H_\chi = (\lambda - \mu) A_\chi B_\chi.$$

Under admissibility ($A_\chi \neq 0$, $\lambda \neq \mu$) the rank-drop $\det H_\chi = 0$ is exactly the scalar L -zero closure $L(\frac{1}{2} + i \log Z, \chi) = 0$.

Theorem 16.4 (Gram rank-drop is the same event; `gram_rank_drop_iff_det_cell_zero`, `gramH_rank_drop_iff_L_zero`, `helix_eigenheight_real`). For any cell pencil L^c the Gram matrix $G^c = (L^c)^H L^c$ is Hermitian positive semidefinite with the same kernel as L^c , and in the square case $\det G^c = |\det L^c|^2$; so the Gram rank-drop $\det G_{\chi,h}^c = 0$ is again exactly the L -zero closure (`gramH_rank_drop_iff_L_zero`). The generalized eigenheights of such a positive pencil are real Rayleigh quotients $Z = \langle v, Av \rangle / \langle v, Bv \rangle$ (`helix_eigenheight_real`, `helix_eigenheight_rayleigh`), the no-log-trap height diagonal being $Z_n = n$, not $\log n$.

16.3 Admissibility of the source-height readout

Theorem 16.5 (Real source-height $\Rightarrow$ critical line; `Admissible.admissible_iff_re_half`, `admissible_real_height_implies_critical_readout`, `nontrivial_zero_represented`). A complex point is the readout `criticalReadout(Z)` of a real source-height Z iff it lies on the critical line $\operatorname{Re} = \frac{1}{2}$ (`admissible_iff_re_half`); the backward direction realizes it at $Z = e^{\operatorname{Im} \rho}$. Equivalently an off-line ρ admits no real source-height (`offline_not_admissible`): the height solving the readout is real iff $\operatorname{Re} \rho = \frac{1}{2}$ (`height_im`, its imaginary part being $-(\operatorname{Re} \rho - \frac{1}{2})$). Consequently every nontrivial critical-line zero is represented by an admissible real source-height Gram eigen-event $Z = e^t > 0$ (`nontrivial_zero_represented`).

Remark 16.6 (Projection primacy — the completeness question; `ProjectionPrimacy`, `projectionPrimacy_iff_exhausts`, `zero_source_admissibility_of_projectionPrimacy`). The readout reaches exactly the critical line for real Z (`criticalReadout_re`), so the 3D carrier's projection cannot land off-axis and cannot host an off-line zero; Theorem 16.5 represents every zero presented on the line. The projection-primacy principle (`ProjectionPrimacy`) asserts that this projection is complete — that its real-height readout exhausts the nontrivial zeros — and is proved equivalent to that exhaustion (`projectionPrimacy_iff_exhausts`), hence to GRH for χ ; conditional on it, every nontrivial zero (with no on-line hypothesis) is a represented admissible eigen-event (`zero_source_admissibility_of_projectionPrimacy`, `RH_of_everyZeroAdmissible`). The same principle is the representability bridge `EveryZeroAdmissible`; these two are the formalized faces of the one undecided question (Remark 17.7). The de Branges IsHB Λ condition is the same RH content in the classical de Branges form (Remark 15.10), which this work connects to but neither formalizes nor relies on — open work in the de Branges program, not a conditional of this paper. The open question itself is posed, never assumed: RH fails only if a factor outside the 3D model produces an off-line zero, which the carrier provably cannot. As an unconditional complement, the perturbation (Rouché-type) stability of the focal zeros is fully proved: at a simple critical-line zero γ where the L -channel dominates the bridge error there is a unique nearby focal zero, with the explicit location bound $|T_d - \gamma| \leq 2\varepsilon/a$ (`real_simple_zero_location`, `real_simple_zero_exists`, `real_simple_zero_unique`).

17 Admissibility: the carrier has no native off-axis support

The constructions above combine into a single intrinsic, on-axis statement about the carrier's *state space* — sharper than “a phasor plot that finds the zeros”: it reframes the critical line as the carrier's admissible state space, the object whose completeness is projection primacy (Remark 17.7). A point of the vertical line $s = \sigma + iy$ rides the carrier as a stable, scale-normalized fiber only when its amplitude $n^{-\sigma}$ reciprocates the emergent area-law radius $r_n \sim \sqrt{n}$ to a finite positive scale. We call such a σ an *admissible (scale-balanced) state*.

Definition 17.1 (Scale-balanced admissible state; `ScaleBalanced`). *For a carrier (p, r) and abscissa $\sigma \in \mathbb{R}$, σ is scale-balanced when the scale-balance product has a positive finite limit: $\exists L > 0$, $n^{-\sigma} r_n \rightarrow L$, where $r_n = \text{carrierRadius}(p, r, n)$ is the emergent area-law radius (Definition 3.8, Theorem 3.9).*

Theorem 17.2 (The admissible exponent is forced, not chosen; `scaleBalanced_iff`, `scaleBalanced_setOf_eq`). *For every carrier (p, r) with $r > 0$, σ is scale-balanced iff $\sigma = \frac{1}{2}$; equivalently $\{\sigma \in \mathbb{R} : \sigma \text{ scale-balanced}\} = \{\frac{1}{2}\}$. The critical abscissa is the unique exponent compatible with the arclength area law (Theorem 3.11), and the conclusion is gauge-independent: the gauge (p, r) fixes only the value of the limit, never which σ is admissible.*

Corollary 17.3 (No native off-axis support; `not_scaleBalanced_of_ne`). *For $\sigma \neq \frac{1}{2}$ the fiber is not a scale-balanced state: its amplitude $n^{-\sigma}$ is not scale-normalized to the carrier (it decays faster than the radius grows, or outruns it). The carrier supports no admissible off-axis state.*

Remark 17.4 (Answering the “designed around $\frac{1}{2}$ ” objection). *The exclusion of off-axis states is not built in: $\frac{1}{2}$ is derived from the arclength area law (Theorems 3.9, 3.11) as the unique exponent at which $n^{-\sigma}$ reciprocates $r_n \sim \sqrt{n}$. It is the $\frac{1}{2}$ of $\sqrt{n} = n^{1/2}$, not a posited abscissa.*

The functional equation supplies the global counterpart: the reflection $s \mapsto 1 - s$, which pairs the two chiral fibers, fixes exactly the critical line.

Theorem 17.5 (The reflection fixes exactly the critical line; `reflection_fixes_iff`). *For $s \in \mathbb{C}$, $\text{Re}(1 - s) = \text{Re } s \iff \text{Re } s = \frac{1}{2}$. Off-axis, $s \mapsto 1 - s$ sends an abscissa imbalance σ to the opposite imbalance $1 - \sigma$ — a dual pair, not a self-conjugate state — so the double-ended carrier closes only on the fixed locus $\text{Re } s = \frac{1}{2}$.*

Theorem 17.6 (The hierarchy, assembled; `no_native_offaxis_support`). *For every carrier (p, r) with $r > 0$, height y , and site n , the following hold simultaneously:*

- (1) *the area law $r_n^2/n \rightarrow r(\pi/3)/\pi$ (so $r_n \sim \sqrt{n}$), from the arclength geometry;*
- (2) *scale balance holds iff $\sigma = \frac{1}{2}$ (forced, not chosen);*
- (3) *every $\sigma \neq \frac{1}{2}$ is inadmissible;*
- (4) *at the balanced axis the chiral fibers $z, \bar{z}$ form a determinant-one unitary transverse block (Theorem 12.4); and*
- (5) *$s \mapsto 1 - s$ fixes exactly $\text{Re } s = \frac{1}{2}$, on which the de Branges variable is real (Corollary 15.9).*

The chain

$$r_n \sim \sqrt{n} \implies \sigma = \frac{1}{2} \implies \text{unitary chiral block} \implies \det = 1 \implies s \leftrightarrow 1 - s \text{ closes on } \text{Re } s = \frac{1}{2}$$

is the carrier's intrinsic logic: it reads as a geometry, not a graphing of $L(\frac{1}{2} + it)$.

Remark 17.7 (Projection primacy: the pivotal hypothesis). *Theorem 17.6 establishes that the carrier’s admissible state space — the image of its real-height projection $Z \mapsto \frac{1}{2} + i\tau_\chi(\log Z)$ — is the critical line: off-axis points are unsupported / inadmissible / not scale-normalized, so the 3D representation cannot host an off-line zero (`criticalReadout_re`). Whether the projection is also complete — whether it exhausts every nontrivial zero — is the projection-primacy principle (`ProjectionPrimacy`), proved equivalent to the real-height exhaustion of the zeros (`projectionPrimacy_iff_exhausts`) and hence to GRH for χ ; conditional on it the exhaustion is unconditional (`zero_source_admissibility_of_projectionPrimacy`). This is the open consideration, and it is a genuine dichotomy: either the projection is primary — the more accurate 3D model defines the 1D/2D readout completely, and there are no off-line zeros — or some factor outside the 3D representation produces an off-line zero, one the carrier provably cannot. The Riemann Hypothesis is thus not something weaker that this work falls short of, but exactly the completeness of the projection; the work poses the dichotomy and proves the off-axis branch impossible within the model. “Frobenius” here means precisely the prime-multiplication similitude of Definition 12.1.*

18 Formalization

Every definition and theorem above is formalized in Lean 4 [17] against Mathlib [16], in the package `RequestProject` (files `ClosedForm.lean`, `HelixLogFreeFTA.lean`, `HelixCollapseReality.lean`, `LFunctionPhasor.lean`, `GeometricProjectionHolds.lean`, `Faithfulness.lean`, `AreaLaw.lean`, `UnconditionalFrobenius.lean`, `DeBranges.lean`, `FrobeniusSimilitude.lean`, `PerHeightConvergence.lean`, `HeightGrowthActive.lean`, `CircleMonodromy.lean`, `CriticalLineBridge.lean`, `GeometricPhasorClosure.lean`, `Phasor3D.lean`, `FocalEigenheight.lean`, `HarmonicPencilCell.lean`, `AdmissibleEigenheight.lean`, `FullFiber.lean`, `PrimePowerProbe.lean`, `CupIdentity.lean`, `ChiralCup.lean`, `CrossingInduction.lean`, `CrossingEncodingAudit.lean`, `HelixPolarization.lean`, `ConditionalToUnconditional.lean`, `SpectralFiberIsLFunction.lean`, `SpectralZeroKernel.lean`, `CancellationEquivalence.lean`, `SelfAdjointGeneratorReadout.lean`, `SpectralSignFlip.lean`, `NoDoubleCancellation.lean`, `SpectralExhaustion.lean`, `SpectralGramEvent.lean`, `SpectralRealizationGRH.lean`, `FocalResidualVanishes.lean`, `FocalCancellationFindsZeros.lean`, `FocalCancellationEigenstate.lean`, `GeometricReadout.lean`, and `MeromorphicTraceIdentity.lean`). Each named theorem compiles with no `sorry`; the kernel axiom footprint reported by `#print axioms` (equivalently `lean.verify`) is

{ `propext`, `Classical.choice`, `Quot.sound` },

the three standard Mathlib axioms, with no construction-specific axiom. The correspondence between the statements here and the Lean declarations is given by the identifiers in parentheses throughout.

The capstone identities — the strip extension (Theorems 7.1, 7.2, 7.3), the conjugacy of the two chiralities (Theorem 11.1), and the determinant identity (Theorem 12.3) — are collected in `Faithfulness.lean`; the geometry (Section 3) and the log-free winding (Section 4) live in `ClosedForm.lean` and `HelixLogFreeFTA.lean`, with the emergent area-law radius (Theorem 3.9, Definition 3.8) and the scale-critical exponent (Theorem 3.11) in `AreaLaw.lean`. The self-adjoint eigenstate design (Section 12.1) lives in `UnconditionalFrobenius.lean` and the Hermite–Biehler / de Branges framework (Section 15) in `DeBranges.lean`, with their connection to the carrier — the eigenstate identification (`spin_eq_spectralWave`, `frobenius_eigenstate_det_one`) and the on-line de Branges evaluation (`criticalLine_deBranges_real`, `zeroHarmonic_deBranges_real`) — collected in `Faithfulness.lean`. The compact local spectral mechanism of Section 12

— the Frobenius similitude (Definition 12.1, Theorem 12.2), the unitary transverse block (Theorem 12.4), the eigenstate–data orthogonality (Proposition 12.8), and the admissibility / no-off-axis-support results of Section 17 (Theorems 17.2, 17.5, 17.6) — lives in `FrobeniusSimilitude.lean`, assembled in `frobenius_local_spectral_mechanism` and `no_native_offaxis_support`. The remaining layers map as follows: the per-height convergence on the line (Section 7.1) is `PerHeightConvergence.lean`; the constant height law and active fraction (Remark 3.2) and the exact $\pi/3$ monodromy (Theorem 3.15) are `HeightGrowthActive.lean` and `CircleMonodromy.lean`; the geometric phasor-closure dictionary and lane split (Section 7) are `GeometricPhasorClosure.lean` and `CriticalLineBridge.lean`; the three-channel 3D phasor (Section 6) is `Phasor3D.lean`; and the harmonic pencil, focal eigenheight, and source-height admissibility (Section 16) are `HarmonicPencilCell.lean`, `FocalEigenheight.lean`, and `AdmissibleEigenheight.lean`. The chiral cup and its Hilbert completion (Section 13) live in `CupIdentity.lean`, `ChiralCup.lean`, `CrossingInduction.lean` (with the non-vacuity audit `CrossingEncodingAudit.lean`) and `PrimePowerProbe.lean`; the Kähler polarization, the trace formula, and the cohomological re-expression of the zeros (Section 14) are `HelixPolarization.lean`; the convergent η -fibre (Section 7.2) is `FullFiber.lean`; and the unconditional Paley–Wiener template (Theorem 15.7) is `ConditionalToUnconditional.lean`.

The updated spectral/focal layer is split into smaller audit modules. `SpectralFiberIsLFunction.lean` proves that the 3D spectral-fiber readout converges to the analytic L -function (`spectral_fiber_is_Lfunction`) and that the eta-configured principal readout recovers ζ on the punctured strip (`etaSpectralFiber_readout_recovers_zeta`). `SpectralZeroKernel.lean` and `CancellationEquivalence.lean` package zeros as kernel events of the configured spectral operator (`spectral_zero_has_kernel`, `spectral_kernel_iff_zero`, `cancellation_equivalence`). The self-adjoint readout and its Gram markers live in `SelfAdjointGeneratorReadout.lean`, `SpectralSignFlip.lean`, `NoDoubleCancellation.lean`, `SpectralExhaustion.lean`, and `SpectralGramEvent.lean`: these record the spectral height `specHeight`, the signed mode `specBchan`, the rank-drop criterion `specGram_det_zero_iff`, sign-change and real-axis facts (`spectral_cancellation_sign_flip`, `sign_flip_only_on_real_axis`), distinct-mode exclusion (`no_double_cancellation`), and the critical-line implication from an L -zero to a harmonic Gram event (`L_zero_imp_spectral_gram_event`). The focal-cancellation modules `FocalResidualVanishes.lean`, `FocalCancellationFindsZeros.lean`, and `FocalCancellationEigenstate.lean` formalize that the normalized focal residual vanishes exactly with the scalar L -readout (`focal_residual_zero_iff_L_zero`), that an on-line zero at ordinate y is found at source height e^y (`focal_cancellation_finds_online_zero`), and that an exact charged focal cancellation produces a standing/eigenstate package with conserved magnitude (`focal_cancellation_produces_eigenstate`). `GeometricReadout.lean` collects the height-readout and no-double-residue facts, and `MeromorphicTraceIdentity.lean` records the completed trace identities for the spectral operator.

Remark 18.1 (Scope). *This paper presents a geometric state space: the double-ended Frobenius helix carrier and its phasor fiber, whose admissible scale-balanced states exist only on the critical line (Section 17) and on which the vanishing points are exactly the on-line zeros of ζ and every Dirichlet L -function, by the formalized equivalence (Theorem 10.1, Section 7.2). Its central contribution is intrinsic: the carrier has no native off-axis support (Theorem 17.6), and RH is reframed as the projection-primacy principle — the completeness of the carrier’s real-height projection (`projectionPrimacy_iff_exhausts`), equivalent to GRH, with the off-axis branch proved impossible inside the model. Its mathematical content is exactly the statements proved above — the carrier, the area-law derivation of the critical exponent (Theorem 3.11), the accumulation to $L(s, \chi)$ on $\text{Re } s > 0$, the explicit-formula reading of the zeros as poles, the reality of the completed object*

on the critical line, the conjugacy of the two chiralities, the Frobenius similitude (Theorem 12.2), the unitary transverse block (Theorem 12.4), the eigenstate–data orthogonality (Proposition 12.8), the self-adjoint eigenstate realization of the chiral eigenphases (Section 12.1), the de Branges evaluation placing the on-line cancellations as real spectral points (Section 15), and the admissibility restriction of the carrier’s state space to the critical line (Section 17). The pivotal hypothesis is projection primacy — the completeness of the 3D-to-1D projection (Remark 17.7) — formalized in two guises: that the real-height readout exhausts the zeros (`projectionPrimacy_iff_exhausts`), and that a vanishing produces the eigenstate (Remark 12.11). The classical de Branges IsHB Λ condition (Remark 15.10) is the same RH content in the de Branges form — connected here but neither formalized nor relied on (open work, not a conditional of this paper). This work does not decide projection primacy: it poses the dichotomy — the projection is primary, so there are no off-line zeros, or an unmodeled factor produces one — and proves that the off-axis branch cannot arise within the 3D carrier. The Riemann Hypothesis is exactly the resolution of this dichotomy, which this work frames but neither assumes nor proves.

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